

Sagor Das

BIOMEDICAL ENGINEERING STUDENT & RESEARCHER



CONTACT

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ABOUT MYSELF

Biomedical Engineering student at Military Institute of Science and Technology (MIST) with hands-on research experience at the National Institute of Biotechnology (NIB). Skilled in molecular biology, microbial biotechnology, bioinformatics, and computational drug discovery. Experienced in microfluidics, finite element analysis (ANSYS, COMSOL), biomechanical modeling (OpenSim), and AI-based healthcare systems. Actively engaged in interdisciplinary biomedical research combining wet lab and computational approaches. Seeking opportunities in biomedical research, diagnostics, and healthcare innovation.

Education

- **Bachelor of Science (B.Sc.)** in Biomedical Engineering
Military Institute of Science and Technology
Year: 2026
Relevant Courses: Molecular Biology, Bioinformatics, Medical Instrumentation, Biomaterials, Signal Processing, Microfluidics, Biomechanics
Result: CGPA 3.66 out of 4
- **Higher Secondary Certificate (HSC) — Science**
Subjects included Physics, Chemistry, and Biology
Bhasha Soinik Ajit Guha College, Comilla
Year: 2021
Result: GPA 5.00 out of 5
- **Secondary School Certificate (SSC) — Science**
Subjects included Physics, Chemistry, and Biology
Year: 2019
Result: GPA 5.00 out of 5

PROFESSIONAL EXPERIENCE

Research Assistant

- MIST (Military Institute of Science and Technology)
Duration: May 2026 – Present

EXPERIENCE

Internship Trainee

- **National Institute of Biotechnology**
Duration: May 12, 2025 – June 4, 2025

Teaching Experience

- Bioinformatics for Drug Design and Development
- MIST Einthoven Club

RESEARCH INTERESTS

- Prosthetic hand and leg development
- Medical Robotics & Physical Systems
- Bio-Materials & Microfluidics
- AI in Healthcare and Rehabilitation
- Biosignal and Image Processing
- Computational Biology
- Soft Matter & Bioinspired Robotics
- Point-of-Care Technologies & Wearables

B.Sc. THESIS

Development of a Microfluidic Device for Detection of Anti-dsDNA Autoantibodies and Therapeutic Screening

Status: Complete

- Designed a microfluidic biosensor for detection of anti-dsDNA autoantibodies, a key biomarker for Systemic Lupus Erythematosus (SLE)
- Integrated computational drug screening and binding affinity prediction methods
- Focused on early-stage diagnosis of autoimmune diseases using point-of-care technology

PROJECTS

1. AI-Powered Web-Based Rehabilitation Exercise Tracker (2025–2026)

- Built AI-based rehabilitation tracking system using pose and hand detection
- **Tools:** HTML, CSS, JS, PYTHON

2. Digital Heart Murmur Detection System (2024)

- Developed signal processing-based stethoscope system
- **Tools:** ESP32, embedded system integration

3. Brain Tumor Segmentation Using Active Contour (2023)

- Implemented active contour (snake model) for biomedical image segmentation.
- **Tools:** MATLAB

4. Andromorphic Robotic Arm (EMG & Remote Control)

- Developed a dual-mode prosthetic robotic arm controlled using EMG signals and a web-based remote interface.
- **Tools:** ESP32, EMG sensors, Arduino ecosystem, web interface (HTML/CSS/JS)

5. Microfluidic Lab-on-Chip System

- Designed a microfluidic lab-on-chip platform to study fluid flow and drug transport across a membrane barrier.
- Fabricated the microfluidic device using PDMS through molding and casting techniques.
- **Tools:** ANSYS

6. Identification of Potential Drug Targets in HTLV-1 via Integrated In Silico Approaches

- Conducted computational analysis to identify potential drug targets in HTLV-1 using integrated in silico methods.
- **Tools:** AutoDock, PyRx, BLAST, Discovery Studio, GROMACS

7. Edible Soft Robotics (2025–2026)

- Developed a soft robotic system using edible and biodegradable materials for sustainable biomedical applications, with potential use in assisting patients with paralysis and those unable to take medication orally.
- Focused on pneumatic actuation, material formulation, and deformation control.
- **Tools:** Gelatin-based soft robot fabricated using molding and casting techniques

PROJECTS

8. Biomechanical Impact of Footwear on Gait and Knee Angle (2024)

- Conducted a biomechanics study to evaluate the effects of different footwear on gait, ground reaction force, center of pressure (COP), and knee joint angle
- Collected and analyzed gait data under multiple footwear conditions including barefoot, flat shoes, sports shoes, boots, and high heels
- **Tools:** Vicon Motion Capture System, Biomechanical Gait Analysis, Force Plate, MATLAB, Data Analysis.

CURRENT PROJECTS

1. Development of a prosthetic leg
2. Design and Development of a Low-Cost Prosthetic Hand with Hybrid EMG and Tactile Control
3. Microfluidic Device for Detection of Anti-dsDNA Autoantibodies
4. Rehabilitation Exercise Tracker version 2.0

WEB DEVELOPMENT PROJECTS

1. Einthoven Club Official Website

- Developed and contributed to the official website of the Biomedical Engineering club at Military Institute of Science and Technology.
- **Link:** <https://einthovenclub.mist.ac.bd/>

2. AI-Powered Web-Based Rehabilitation Exercise Tracker (2025–2026)

- Built AI-based rehabilitation tracking system using pose and hand detection
- Used computer vision and MediaPipe for real-time posture correction
- Designed for remote physiotherapy applications
- **Link:** https://sagorghhj253.github.io/sagor_rehabilitation_project_BME410/

3. Andromorphic Robotic Arm – Remote Control Interface

- Designed a web interface for real-time remote control of a prosthetic robotic arm system.
- Integrated with ESP32-based hardware for dual-mode (EMG + web) control.
- Includes authentication-based access (username(“sagor”)/password(“46”) system).
- **Link:** <https://oishy17.github.io/hand/>

4. Digital Heart Murmur Detection System (2024)

- Developed signal processing-based stethoscope system
- Extracted time-frequency features for classification
- Targeted early detection of valvular heart disease

PUBLICATIONS / CONFERENCE PAPERS

- Breast Cancer Subtype Prediction using KNN with PCA Feature Selection (2025) – Manuscript in preparation
- Identification of Potential Drug Targets in HTLV-1 via Integrated In Silico Approaches (2025) – Manuscript in preparation
- Markerless Real-time Pose Estimation for Home-based Rehabilitation Exercise Tracking and Form Assessment- Manuscript under review
- Brain Tumor Segmentation and 3D Visualization Using Active Contour Models: From Multi-Point to Single Seed Point Initialization -Manuscript under review

Technical Skills

Programming Language

- C Programming
- Python (Basic)
- Linux Command Line
- MATLAB (Signal & Image Processing)
- Web Development: HTML, CSS, JavaScript

Simulation & Computational Tools

- GROMACS (molecular dynamics simulation)
- ANSYS (fluid dynamics and engineering simulation)
- AutoDock /PyRx (molecular docking)
- PyMOL (molecular visualization)
- Discovery Studio (protein-ligand interaction analysis)
- Clustal Omega (sequence alignment)
- BLAST, UniProt, PDB
- Machine Learning (KNN, PCA)
- Computer Vision (Pose Detection)

Simulation & Design Tools

- SolidWorks (3D CAD design and modeling)
- Proteus (circuit design - Basic)
- COMSOL Multiphysics
- OpenSim (Biomechanical Modeling)
- Kimera (3D Scene Reconstruction)

Hardware & Embedded Systems

- ESP32 microcontroller
- Arduino platform
- Biomedical sensors

Data analysis

- SPSS

LABORATORY & TECHNICAL SKILLS

- Embedded systems development and biomedical device integration
- Rehabilitation engineering and assistive technology development
- Biochemistry laboratory techniques
- Biomaterials
- Biomechanics
- Molecular biology
- Biomedical instrumentation and biosignal acquisition
- Signal processing
- Biomedical image processing
- Wet lab and dry lab research methodologies

LEADERSHIP & EXTRACURRICULAR ACTIVITIES

CLUB CO-ORDINATOR

MIST Einthoven Club

Year: 2026

- Organized a successful biomedical-focused event, “Celestra ’26”, promoting innovation and interdisciplinary collaboration in biomedical engineering.

CERTIFICATIONS & TRAININGS

- **Champion, Poster Presentation Competition 2025**
 - Military Institute of Science and Technology presented the poster “Andromorphic Robotic Hand,” focusing on EMG and remote control systems
- Hands-on Training in Next-Generation Sequencing (NGS) – National Institute of Biotechnology (2025)
- Bioinformatics and Computational Drug Discovery Workshop – NIB (2025)
- MIST Einthoven Club Presidential Panel Certification CELESTRA’26 (2026)

LANGUAGES

- **Bengali** — Native Proficiency
- **English** — Professional Working Proficiency

REFERENCES

Lt Col Md. Maruf Hasan, Ph.D, EME

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Military Institute of Science and Technology (MIST)

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